Akoizo Original Practice - Form 2

Division B | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Snow depth equals liquid-water depth automatically?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only in mountains

B. Only in spring

C. No

D. Always

Answer: _____

2. (2 points) Dissolved load consists of…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only gravel

B. Only floating wood

C. Only sand

D. Ions and molecules in solution

Answer: _____

3. (2 points) Hydrograph area represents…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Volume

B. Only peak stage

C. Only velocity

D. Only basin area

Answer: _____

4. (2 points) Every lake mixes twice per year?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if freshwater

B. No

C. Always

D. Only if deep

Answer: _____

5. (2 points) A contour connects equal…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Rainfall always

B. Chemical concentration always

C. Elevation

D. Discharge

Answer: _____

6. (2 points) A one-event lab predicts climate?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. No

B. Always

C. Only with more sliders

D. Only for wet regions

Answer: _____

7. (2 points) Competence describes…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only rainfall

B. Largest transportable particles

C. Total water volume

D. Only sediment color

Answer: _____

8. (2 points) A 100-year label means exact spacing?

A. Only for rivers

B. Only after one flood

C. No

D. Always

Answer: _____

9. (2 points) A bog is typically associated with…

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only seawater tides

B. Only bedrock magma

C. Only industrial pipes

D. Precipitation-dominated water supply

Answer: _____

10. (2 points) Vertical exaggeration changes…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Displayed slope appearance

B. Actual geology automatically

C. Measured head automatically

D. Water mass

Answer: _____

11. (2 points) Equal annual rain implies equal reliable supply?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only near oceans

B. Only for groundwater

C. No

D. Always

Answer: _____

12. (2 points) Equal discharge guarantees equal velocity?

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Always

B. Only for rivers

C. Only at flood stage

D. No

Answer: _____

13. (2 points) Stage and discharge are…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Related but different quantities

B. Identical units

C. Always equal numbers

D. Unrelated by any model

Answer: _____

14. (2 points) Wetlands remove all contaminants forever?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if large

B. No

C. Always

D. Only in summer

Answer: _____

15. (2 points) Best monitoring design begins with…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Only convenience

B. Only an attractive map

C. A defined question

D. Maximum instruments

Answer: _____

16. (4 points) Why can runoff lag snowfall?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

Answer: _____
_____
_____

17. (4 points) Why can a sediment-starved flow erode?

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

Answer: _____
_____
_____

18. (4 points) Why can a rating curve change?

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

Answer: _____
_____
_____